

Food and Agriculture

ENGR49580D Technology, Society and Sustainability

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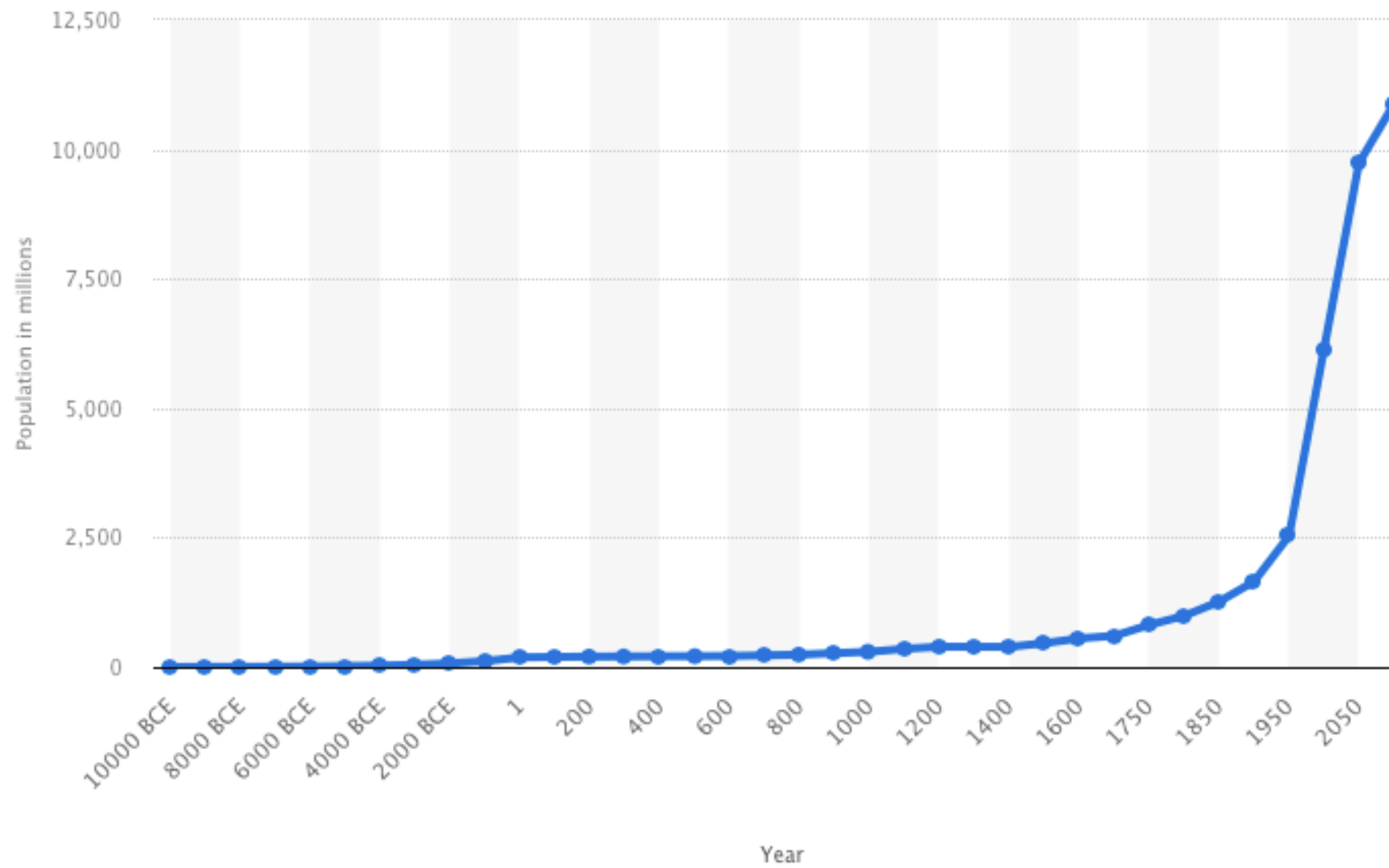
**Corn is a necessary, silver is only a
superfluity.**

Adam Smith, *Wealth of Nations*, I:XI.e, p.210 1776

- The historical role of food
- Technology in agriculture
 - The industrial revolution
 - The green revolution
- Technology in food
 - Genetically modified foods
 - Gene patents

The Historical Role of Food

- For much of human history, human communities lived as either hunter gatherers or subsistence farmers
- The move from hunting and gathering to farming improved food security
- In both models populations were limited by available food resources



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Technology in Agriculture

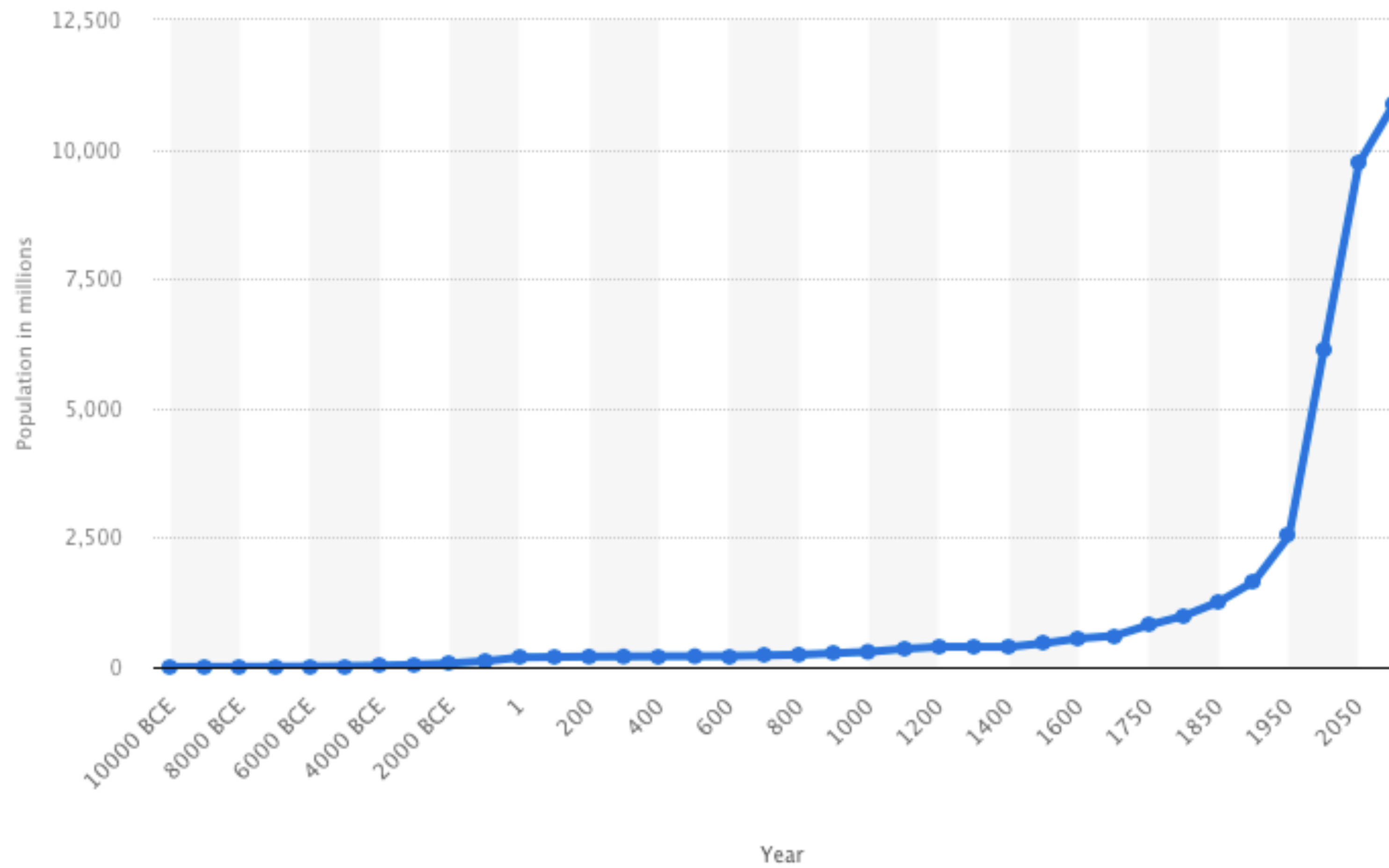
The Industrial Revolution

- The first stage of the industrial revolution was the mechanization of agriculture from 1750 to 1850
- This saw mechanical implements such as
 - The Steam Engine (1698)
 - Adapted for agricultural use (1812)
 - The Cotton Gin (pre industrial) and Spinning Jenny (1774)
 - Threshing Machine (1786)
 - The Mechanical Reaper (1831)
 - The Combine Harvester (1836)

Technology in Agriculture

The Industrial Revolution

- This change in agriculture saw a significant decrease in the number of workers needed to produce food
 - This led to social unrest (1830 Swing riots)
 - This freed farm workers to move to cities to pursue factory work
- The industrialization of agriculture did not significantly alter the fertility (and thus productivity) of land only the labour force required to extract that produce



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The Green Revolution

- The 'Green Revolution' refers to a group of technologies in agriculture aimed at significantly increasing crop yields
 - Plant genetics
 - Modern irrigation systems
 - Chemical fertilizers
 - Pesticides

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The Green Revolution - Origin

- Development of a smaller, fast-growing wheat varietal in Mexico
 - Required less land to produce more grain
 - Efforts supported by
 - Mexican President
 - U.S. government
 - United Nations, the Food and Agriculture Organization (FAO),
 - Rockefeller Foundation

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The Green Revolution - Origin

- Between the 1940s and 1960s Mexico transitioned to agricultural self sufficiency
- International Maize and Wheat Improvement Center (CIMMYT)
 - founded 1943 – Mexico
- International Rice Research Institute (IRRI)
 - founded 1960 – Philippines
 - supported by the Ford Foundation and the Rockefeller Foundation

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The Green Revolution - Dissemination

- Technology Transfers
 - India 1960s
 - Brazil 1960s to present
 - Large scale soil amendment to allow agriculture
- Parallel Developments
 - China 1950s & 60s

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The Green Revolution - Technologies

- High Yield Varieties
 - increased ability to uptake nitrogen
 - dwarf stature

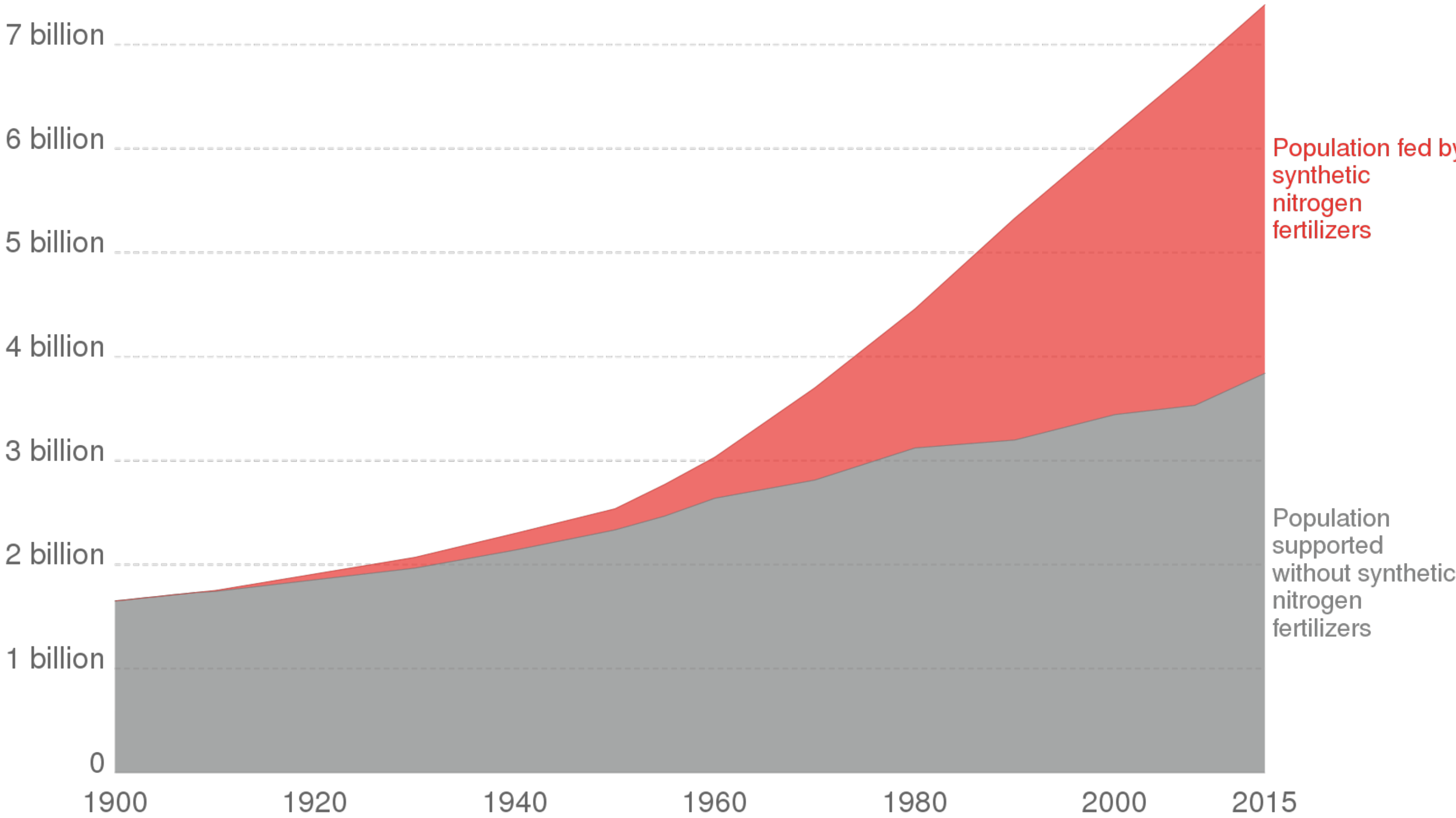
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The Green Revolution - Inputs

- Fertilizers (primarily synthetic nitrogen)
 - Produced from natural gas
- Pesticides
 - Produced from petroleum
- Irrigation
 - Primarily powered by fossil fuels

World population supported by synthetic nitrogen fertilizers

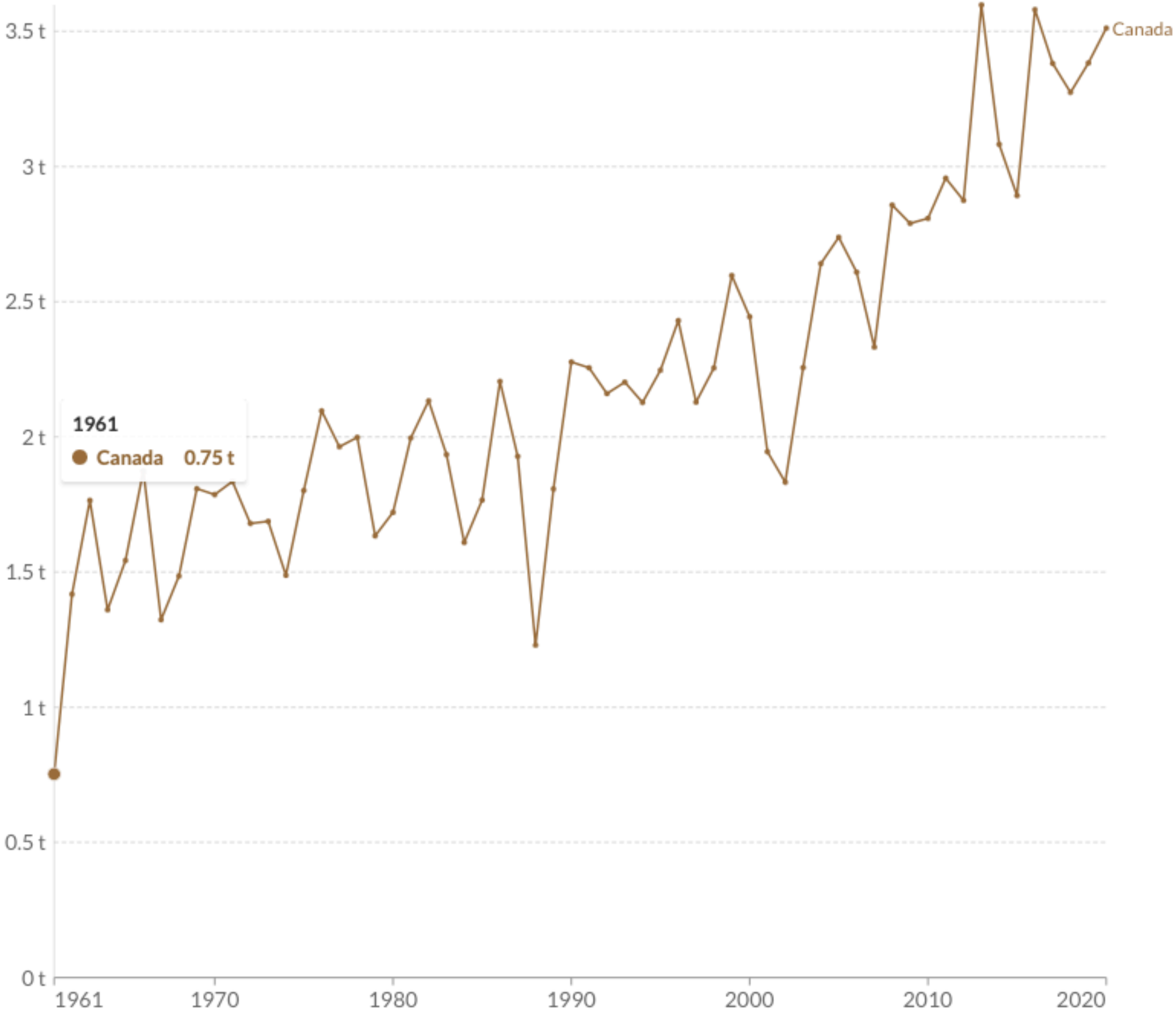
Estimates of the share of the global population which could be supported with and without the production of synthetic nitrogen fertilizers (via the Haber-Bosch process) for food production. Best estimates project that just over half of the global population could be sustained without reactive nitrogen fertilizer derived from the Haber-Bosch process.



Source: Erismann et al. (2008); Smil (2002); Stewart (2005)

Wheat yield

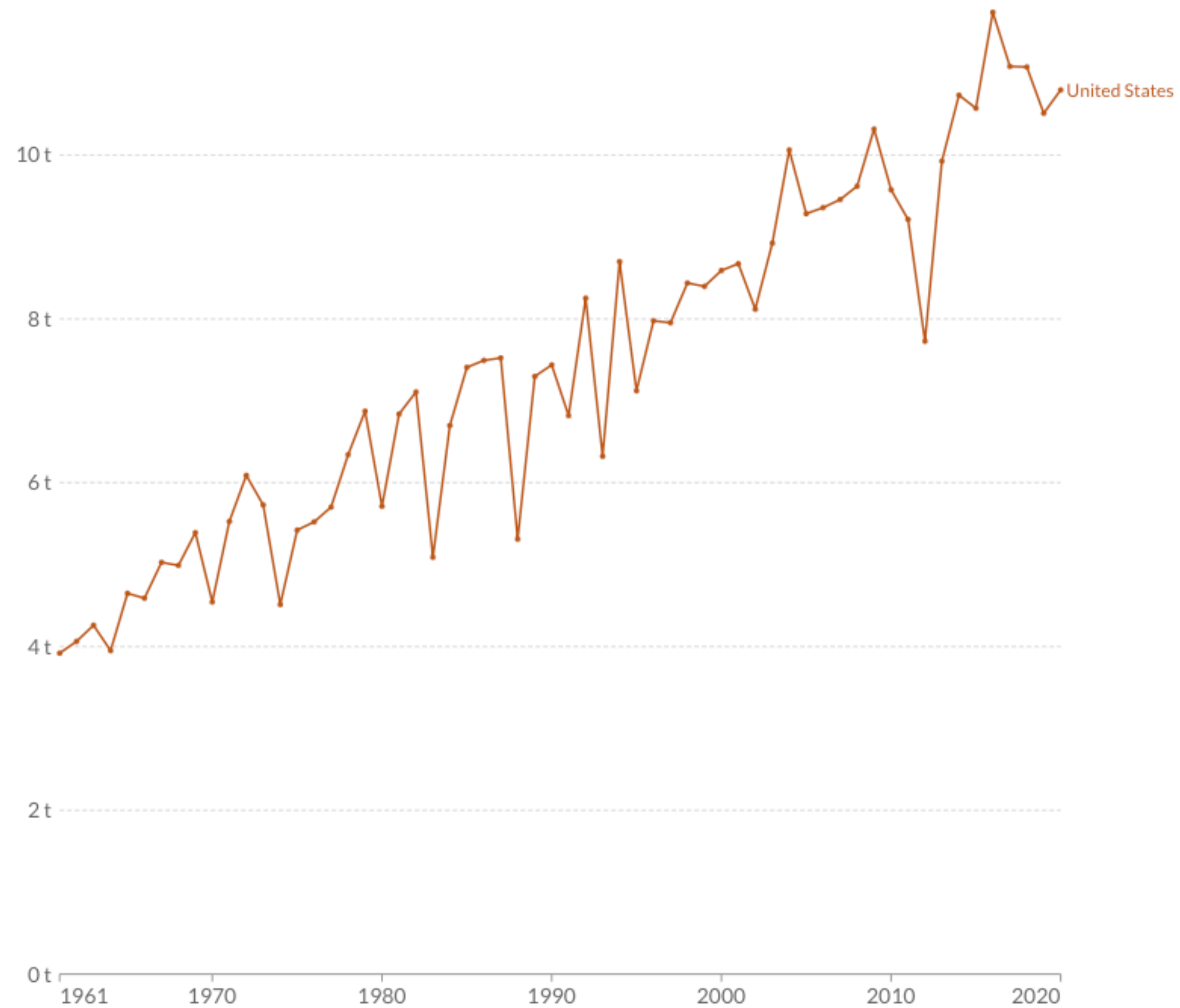
Wheat yields are measured in tonnes per hectare.



Source: UN Food and Agriculture Organization (FAO)

Corn yields

Average corn (maize) yields, measured in tonnes per hectare per year.

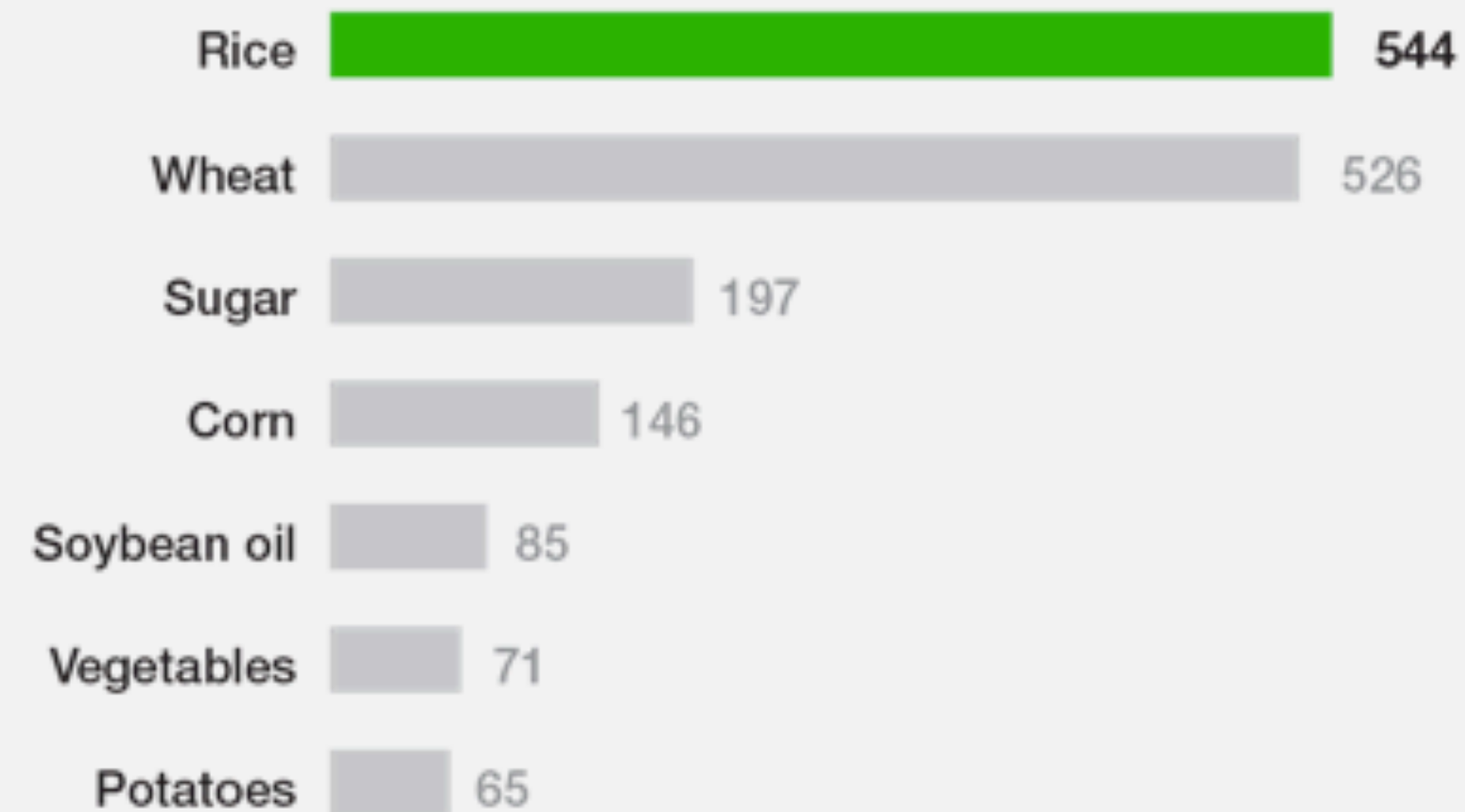


Source: UN Food and Agriculture Organization (FAO)

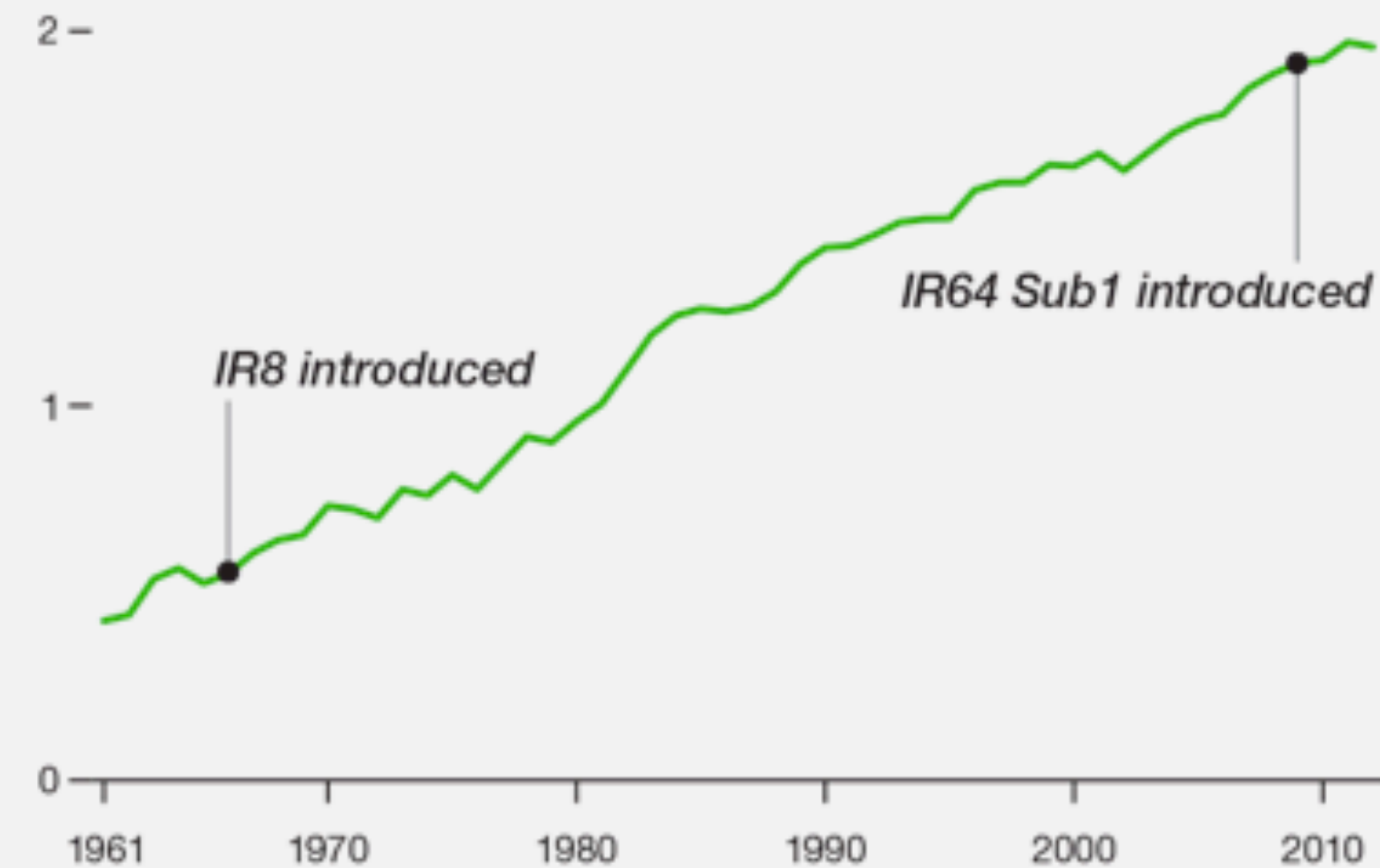
OurWorldInData.org/crop-yields • CC BY

<https://ourworldindata.org/crop-yields>

Global Food Supply *kilocalories per capita*

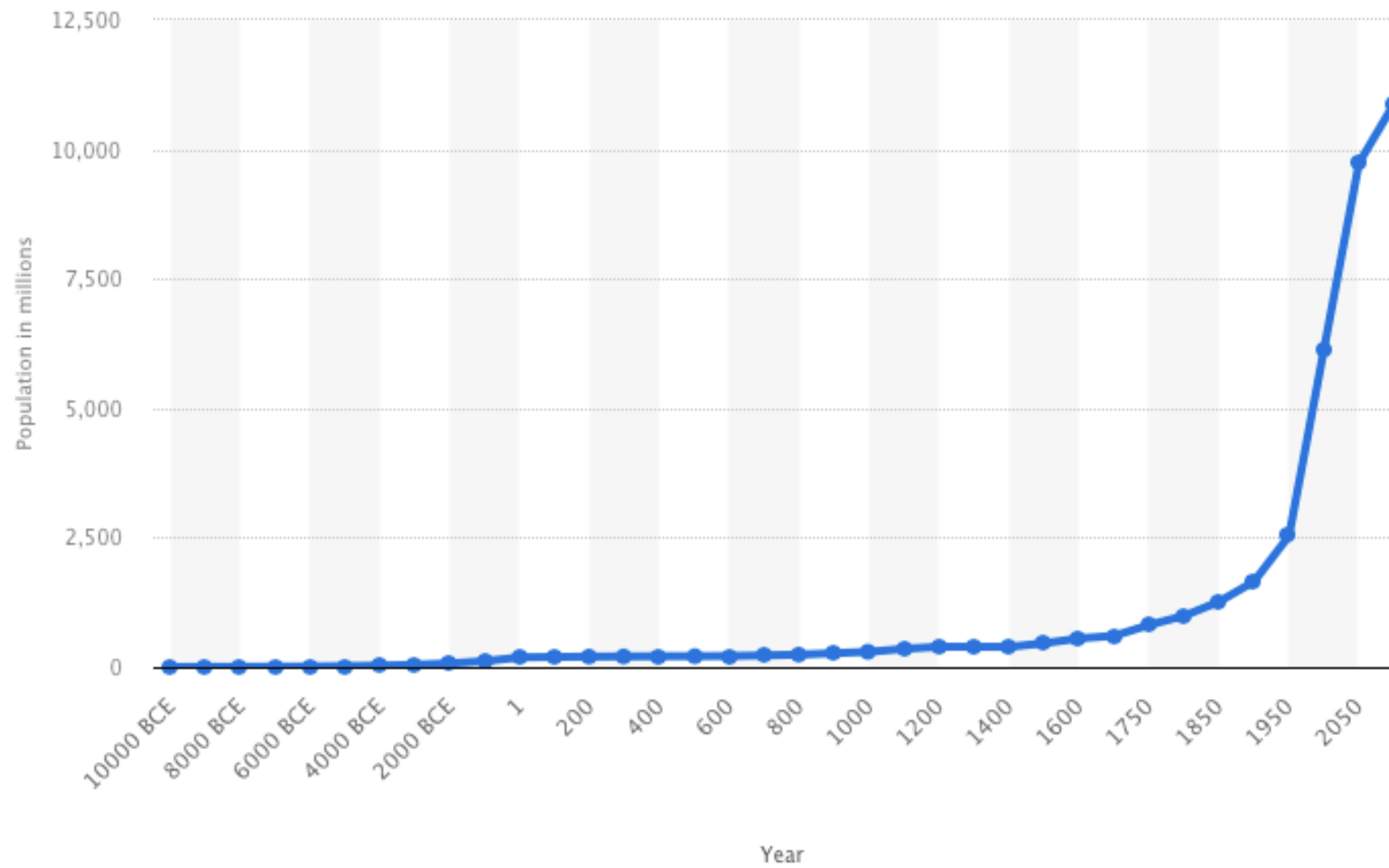


Global Rice Yields *tons per acre*



Rice is the most important food crop in the world, providing more energy to humanity than any other food source. Rice yields have more than tripled since 1961, keeping up with Asia's growing population.

JASON TREAT, NGM STAFF. SOURCES: JANELLE JUNG, IRRI; FAOSTAT



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Genetically Modified Foods



Traditional Breeding

Desired traits are identified in separate individuals of the same species, which are then bred to combine those traits in a new hybrid variety.



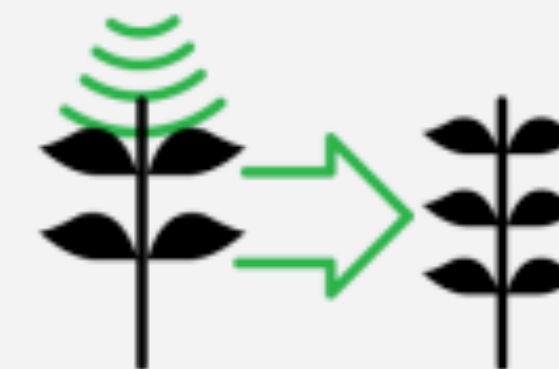
Interspecies Crosses

Breeders can also cross different yet similar species. Modern wheat comes from such hybridizations, some of which happened naturally.



Genetic Modification

Genes identified in one species can be transferred directly to an unrelated species, giving it an entirely new trait—resistance to a pest, say, or to a weed killer.



Mutation Breeding

Seeds are irradiated to promote random mutations in their DNA. If a mutation happens to produce a desirable trait, the plant is selected for further breeding.

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Genetically Modified Foods

- Genetically modified crops are planted on approximately 11% of the world's arable land
 - Primarily used in the US (~50% of cropland)
 - Used since the 1990s
 - Documented to have prevented billions in agricultural losses
 - Resulted in a reduction of pesticide use in the US

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Genetically Modified Foods – Varieties

- Bt corn
 - Contains genes from the bacterium *Bacillus thuringiensis*
 - Wards off corn borers and other pests
- Modified Cotton Varietals planted in China
 - wide-scale reduction in harmful aphids
 - increases in ladybugs and other beneficial insects

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Genetically Modified Foods – Varieties

- Roundup Ready crops
 - Developed by Monsanto (now Bayer, sold to BASF)
 - Corn, Cotton and Soybean varieties tolerate the broad spectrum herbicide Roundup (also manufactured by Monsanto, now BASF)
 - Allows for better weed management
 - Reduced need for tillage and other costly weed management strategies
 - Improved crop yields due to better weed control

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Genetically Modified Foods – Varieties

- Golden Rice
 - Developed by IRRI
 - Contains genes from corn to produce beta-carotene
 - Intended to combat Vitamin A deficiency

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Genetically Modified Foods – Varieties

- Second Generation GM Varietals are being developed with traits that affect the consumer
 - Reduced trans-fat soybeans
 - StayRipe bananas
 - Pharmaceutical producing varieties

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Genetically Modified Foods – Risks

- Gene escape
- Threats to 'heritage' varieties
- Monogenetic crops are vulnerable
- Transfer of allergen producing genes between species

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Genetically Modified Foods – Benefits

- Increased agricultural production
 - Reduced land use
- Reduced pesticide use
- Increased produce shelf life
- Removal of allergens, trans-fats etc.
- Adding of nutrients

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Gene Patents

- Patent authorities have accepted patents on both genes and GMO varieties
- GMO varieties are sold with licences prohibiting self seeding
- Many modern varieties (GMO and traditionally bred) do not produce viable seed
- Farmers must purchase seed each year instead